

Skill7

Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
srivalli@DESKTOP-659T6SD: ~$ cd ~
srivalli@DESKTOP-659T6SD: ~$ mkdir Skill7
srivalli@DESKTOP-659T6SD: ~$ cd Skill7
srivalli@DESKTOP-659T6SD: ~/Skill7$ nano waitpid_demo.c
srivalli@DESKTOP-659T6SD: ~/Skill7$ gcc -Wall -Wextra waitpid_demo.c -o waitpid_demo
srivalli@DESKTOP-659T6SD: ~/Skill7$ ./waitpid_demo
Parent process started. PID = 447
Parent is waiting for child using waitpid()...
Child process started. PID = 448, PPID = 447
Child process completed.
Child terminated normally.
Child exit status = 42
Parent process completed.
srivalli@DESKTOP-659T6SD: ~/Skill7$ nano path_resolver.c
srivalli@DESKTOP-659T6SD: ~/Skill7$ gcc -Wall -Wextra path_resolver.c -o path_resolver
srivalli@DESKTOP-659T6SD: ~/Skill7$ ./path_resolver ls
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/dell/OneDrive/Desktop/openjdk-25/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314:/mnt/c/Users/dell/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dell/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dell/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/dell/AppData/Roaming/npm:/snap/bin

Found: /usr/bin/ls
Executable permission: YES
Command resolved successfully.
```

```
srivalli@DESKTOP-659T6SD: ~$ ./path_resolver gcc
Command resolved successfully.
srivalli@DESKTOP-659T6SD: ~/Skill7$ ./path_resolver gcc
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/dell/OneDrive/Desktop/openjdk-25/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314:/mnt/c/Users/dell/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dell/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dell/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/dell/AppData/Roaming/npm:/snap/bin

Found: /usr/bin/gcc
Executable permission: YES
Command resolved successfully.
srivalli@DESKTOP-659T6SD: ~/Skill7$ ./path_resolver abcxyz
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/dell/OneDrive/Desktop/openjdk-25/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314:/mnt/c/Users/dell/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dell/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dell/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/dell/AppData/Roaming/npm:/snap/bin

Command 'abcxyz' not found in PATH.
srivalli@DESKTOP-659T6SD: ~/Skill7$ ls -l
total 40
-rwxr-xr-x 1 srivalli srivalli 16344 Sep 23 06:25 path_resolver
-rw-r--r-- 1 srivalli srivalli 1294 Sep 23 06:25 path_resolver.c
-rwxr-xr-x 1 srivalli srivalli 16360 Sep 23 06:24 waitpid_demo
-rw-r--r-- 1 srivalli srivalli 884 Sep 23 06:24 waitpid_demo.c
srivalli@DESKTOP-659T6SD: ~/Skill7$ ./waitpid_demo
Parent process started. PID = 476
Parent is waiting for child using waitpid()...
Child process started. PID = 477, PPID = 476
Child process completed.
Child terminated normally.
Child exit status = 42
Parent process completed.
```

```
srivalli@DESKTOP-6S9T6SD:~/Skill7$ ./path_resolver ls
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/Micro
softCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/
c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Use
rs/dell/OneDrive/Desktop/openJdk-25/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/dell/AppData/L
ocal/Programs/Python/Python314/Scripts:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314:/mnt/c/Users/dell/AppData/Local/Microsoft/Win
dowsApps:/mnt/c/Users/dell/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dell/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/dell/AppD
ata/Roaming/npm:/snap/bin

Found: /usr/bin/ls
Executable permission: YES
Command resolved successfully.
srivalli@DESKTOP-6S9T6SD:~/Skill7$ ./path_resolver abcxz
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/Micro
softCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/
c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Use
rs/dell/OneDrive/Desktop/openJdk-25/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/dell/AppData/L
ocal/Programs/Python/Python314/Scripts:/mnt/c/Users/dell/AppData/Local/Programs/Python/Python314:/mnt/c/Users/dell/AppData/Local/Microsoft/Win
dowsApps:/mnt/c/Users/dell/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dell/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/dell/AppD
ata/Roaming/npm:/snap/bin

Command 'abcxyz' not found in PATH.
srivalli@DESKTOP-6S9T6SD:~/Skill7$ |
```